

HOW TO FILL OUT THIS SHEET:



- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

EXAMPLE 1: Debbie Bach

FIRST NAME

D	E	B	B	I	E
A	A	A	A	A	A
H	H			B	B
C	C			C	C
D	D			D	D
E	E			E	E

LAST NAME

B	A	C	H
A		A	A
H	B	H	B
C	C		
D	D		
E			

EXAMPLE 2: Chan Ai Beng

FIRST NAME **LAST NAME**

C	H	A	N
A	A		A
B	B	B	B
C	C		
D	D		
E			

A	I	B	E	N	G
A	A	A	A	A	A
B	B	B	B	B	B
C	C				
D					
E	E	E	E		

EXAMPLE 3: Jamal bin Abas

FIRST NAME

J	A	M	A	L		B	I	N
A								
B								
C								
D								
E								
F								
G								
H								
I								
J								
K								
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W								
X								
Y								
Z								

LAST NAME

A	B	A	S
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T			
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V			
W			
X			
Y			
Z			

FIRST NAME to appear on certificate

[illegible]

LAST NAME to appear on certificate

[illegible]

Are you male or female?
☐ Male ☐ Female



Does anyone in your home usually speak a language other than English?

School name: _____

Town / suburb: _____

Today's date: / / Postcode:

DATE OF BIRTH

Day		Month	Year	
0	0	0	0	0
1	1	1	1	1
2	2		2	2
3	3		3	3
4			4	4
5			5	5
6			6	6
7			7	7
8			8	8
9			9	9

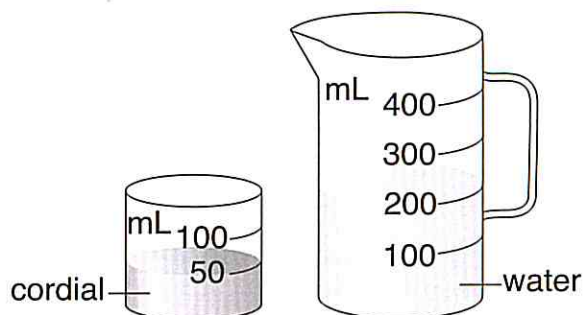
CLASS
(optional)

	(A)	(K)
	(B)	(L)
	(C)	(M)
	(D)	(N)
	(E)	(O)
	(F)	(P)
	(G)	(Q)
	(H)	(R)
	(I)	(S)
	(J)	(T)

TO ANSWER THE QUESTIONS

Example: Ari added cordial to water to make a jug of drink.
What will be the volume of the drink in the jug?

- (A) 50 mL
- (B) 150 mL
- (C) 200 mL
- (D) 250 mL



The answer is 250 mL, so you would fill in the oval **(D)**, as shown.

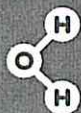
(A) (B) (C) **(D)**



START

- 1 (A) (B) (C) (D)
- 2 (A) (B) (C) (D)
- 3 (A) (B) (C) (D)
- 4 (A) (B) (C) (D)
- 5 (A) (B) (C) (D)
- 6 (A) (B) (C) (D)
- 7 (A) (B) (C) (D)
- 8 (A) (B) (C) (D)
- 9 (A) (B) (C) (D)
- 10 (A) (B) (C) (D)
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- 12 (A) (B) (C) (D)
- 13 (A) (B) (C) (D)
- 14 (A) (B) (C) (D)
- 15 (A) (B) (C) (D)

- 16 (A) (B) (C) (D)
- 17 (A) (B) (C) (D)
- 18 (A) (B) (C) (D)
- 19 (A) (B) (C) (D)
- 20 (A) (B) (C) (D)
- 21 (A) (B) (C) (D)
- 22 (A) (B) (C) (D)
- 23 (A) (B) (C) (D)
- 24 (A) (B) (C) (D)
- 25 (A) (B) (C) (D)
- 26 (A) (B) (C) (D)
- 27 (A) (B) (C) (D)
- 28 (A) (B) (C) (D)
- 29 (A) (B) (C) (D)
- 30 (A) (B) (C) (D)



S

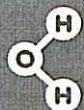
2011 Science Answer Keys

ICAS

International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	A	B	C	B	C	C	A	A	B
2	B	B	D	B	B	D	C	A	A
3	A	A	B	C	C	D	D	C	A
4	C	C	D	D	C	B	C	B	A
5	A	A	A	A	D	C	C	A	D
6	B	C	D	A	C	A	B	C	C
7	C	B	A	D	A	D	C	C	D
8	C	A	D	A	C	C	A	A	C
9	A	B	C	C	B	B	B	C	B
10	D	A	C	D	C	B	B	B	C
11	D	D	D	A	B	B	C	B	D
12	B	B	B	D	D	B	D	C	C
13	C	A	D	B	A	D	A	A	A
14	A	D	C	A	D	C	D	C	C
15	B	A	A	A	B	C	C	B	C
16	D	C	C	C	D	C	C	B	C
17	D	C	C	B	D	D	B	A	C
18	C	D	B	D	C	D	B	A	B
19	A	C	D	D	A	B	C	C	B
20	D	C	D	C	C	C	C	C	C
21	D	B	C	B	C	B	B	D	C
22	C	D	D	C	B	D	C	C	A
23	D	A	A	A	D	A	D	A	C
24	B	A	B	B	A	C	B	B	C
25	C	D	A	D	C	D	D	D	A
26	B	B	C	B	D	D	C	D	B
27	C	A	A	D	A	B	D	B	C
28	D	D	B	B	B	C	D	C	B
29	B	C	D	C	C	D	B	C	C
30	D	D	B	B	C	C	C	D	D

(Please turn over)



S

2011 Science Answer Keys

ICAS
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
31			B	B	B	C	C	C	B
32			A	A	C	B	B	B	D
33			D	C	C	C	B	C	C
34			A	B	C	A	C	B	B
35			A	C	D	C	A	D	C
36			B	B	A	C	B	B	B
37			C	B	A	A	C	C	A
38			D	D	B	B	B	C	D
39			C	B	D	C	B	D	A
40			B	C	B	C	C	C	D
41					C	C	D	C	C
42					C	A	B	D	A
43					C	C	C	C	B
44					D	C	A	D	B
45					C	C	D	D	A

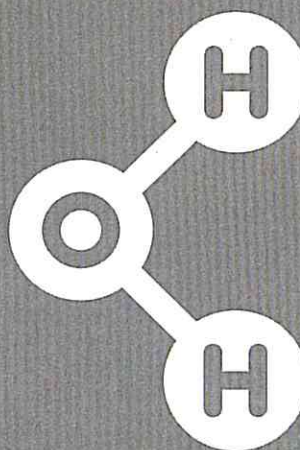
Contact Details

Telephone: (02) 8344 1016
Email: info@eaa.unsw.edu.au
Website: www.eaa.unsw.edu.au



UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

**PAPER
B**



2011
ICAS

International Competitions
and Assessments for Schools

SCIENCE

**Educational
Assessment
Australia**
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

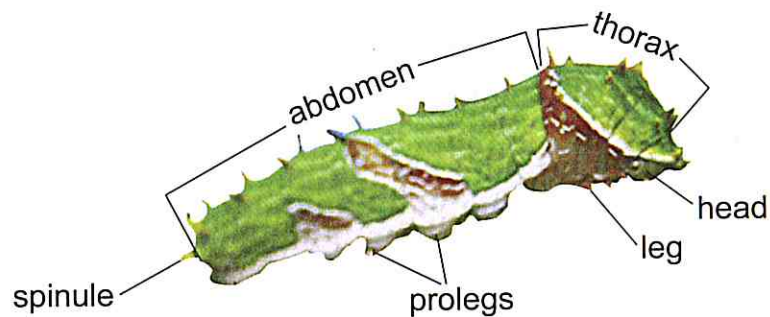
1. This is a photograph of a flower.

What is the width of this flower from X to Y?

- (A) 6 cm
- (B) 7 cm
- (C) 8 cm
- (D) 9 cm



2. This photograph shows an insect larva.

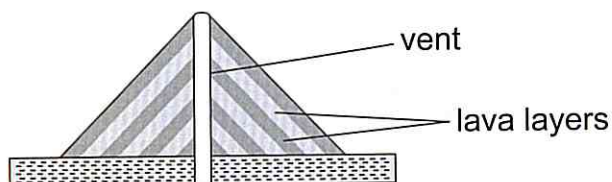


Which part of the body has the prolegs attached to it?

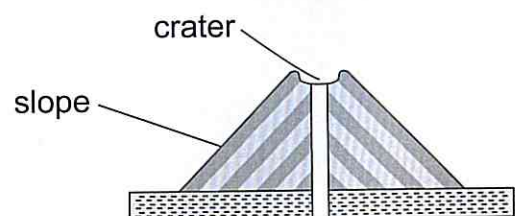
- (A) spinule
- (B) abdomen
- (C) thorax
- (D) head

3. Marty drew these diagrams of two volcanoes.

Volcano 1



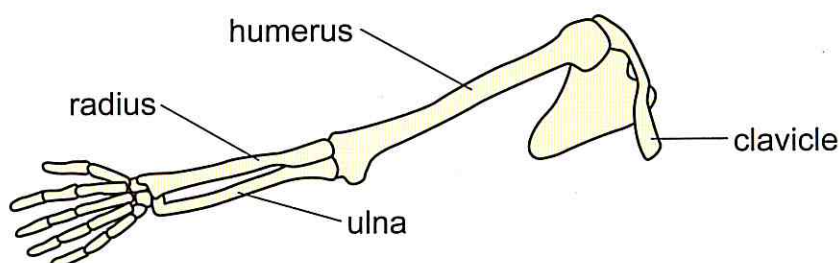
Volcano 2



What is different about these volcanoes?

- (A) the crater at the top
- (B) the location of the vent
- (C) the slope of the sides
- (D) the number of lava layers

4. This drawing shows the bones in a human arm.



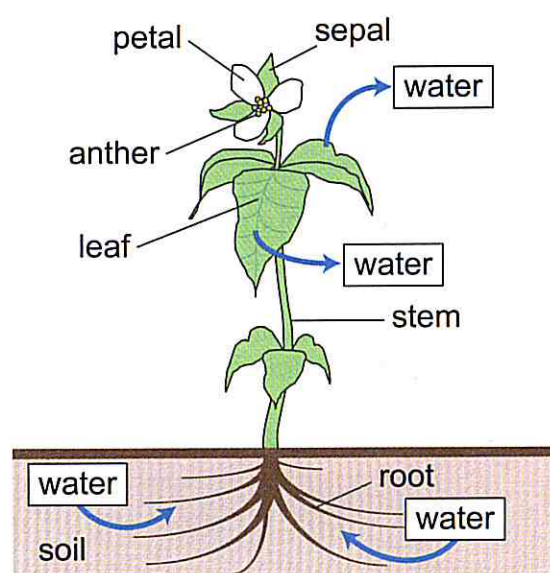
Which is the longest bone in the arm?

- (A) radius (B) ulna (C) humerus (D) clavicle

5. The diagram shows how water can enter and leave a flowering plant.

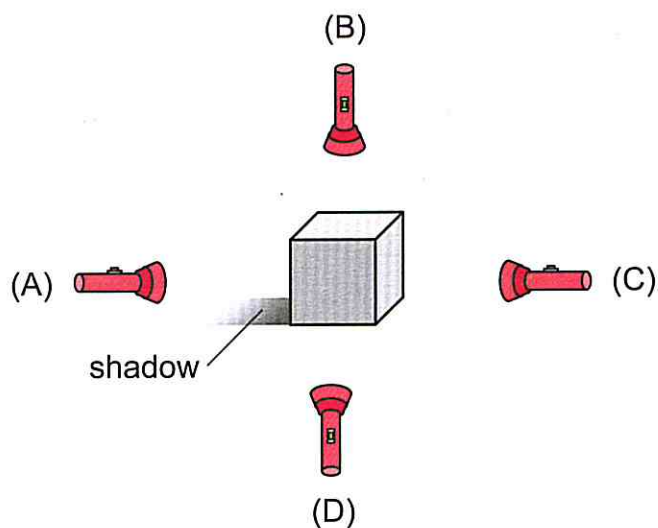
What part of the plant carries water from the roots to the leaves?

- (A) stem
(B) petal
(C) sepal
(D) anther



6. A shadow is formed when an object blocks light.

Which torch would make this shadow when it is switched on?



7. Vertebrates are animals that have a backbone. Warm-blooded animals are able to maintain a relatively constant body temperature in all external environmental conditions. Birds have a backbone and keep their body temperature at a constant level.

	Cold-blooded	Warm-blooded
Vertebrate	group 1	group 2
Non-vertebrate	group 3	group 4

In which group do birds belong?

- (A) group 1
- (B) group 2
- (C) group 3
- (D) group 4

8. Mandy tested four different surf skis and rated them according to speed (how fast they can go) and manoeuvrability (how easily they can turn).

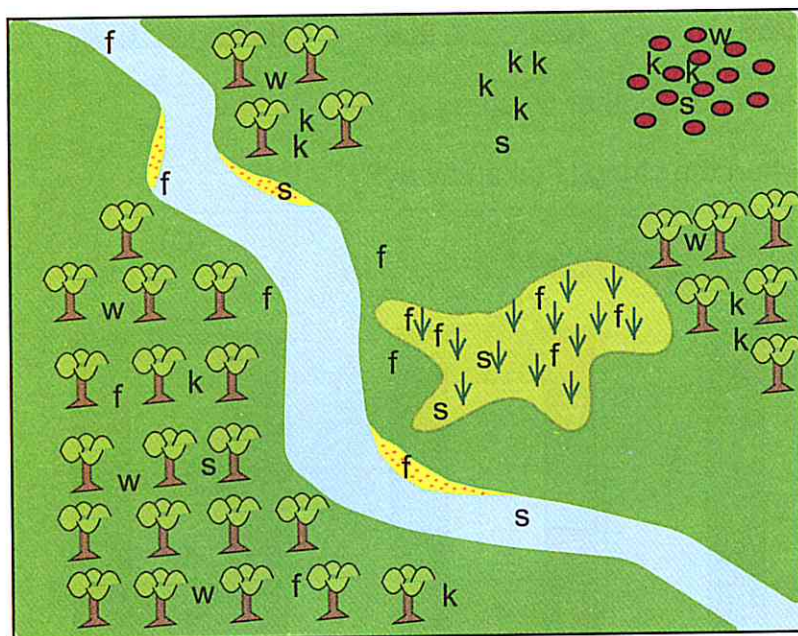
Mandy will be taking part in a long distance race where she will need a surf ski that will go as fast as possible in a straight line.

Which surf ski would best suit Mandy's needs?

	Surf ski	Speed rating (1 = fastest)	Manoeuvrability rating (1 = easiest to turn)
(A)		1	4
(B)		3	2
(C)		2	3
(D)		4	1

9. Jai drew a map of his local area to show where

- particular environments exist
- individual animals have been observed.



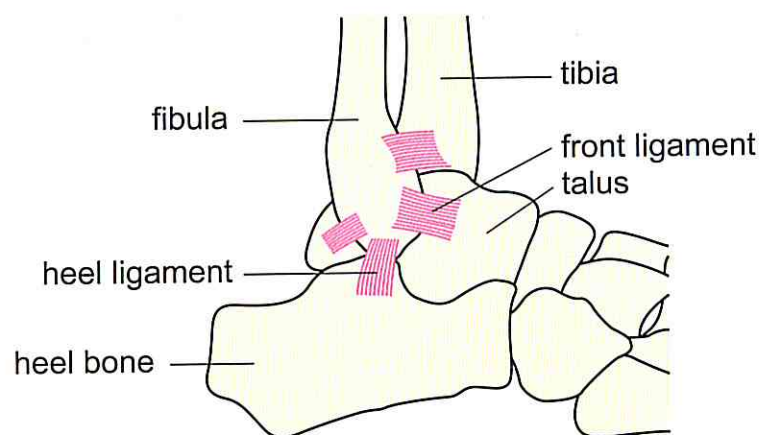
KEY	
Environments	
	grasslands
	swamp lands
	rocky area
	river
	wood lands
	sandy beach
Animals	
f	frog
k	kangaroo
s	snake
w	wombat

Which animal has been observed in all six environments in Jai's local area?

- (A) frog (B) snake (C) wombat (D) kangaroo

10. Ligaments hold bones together.

The diagram shows some ligaments and bones of the ankle.



Which two bones are connected by the front ligament?

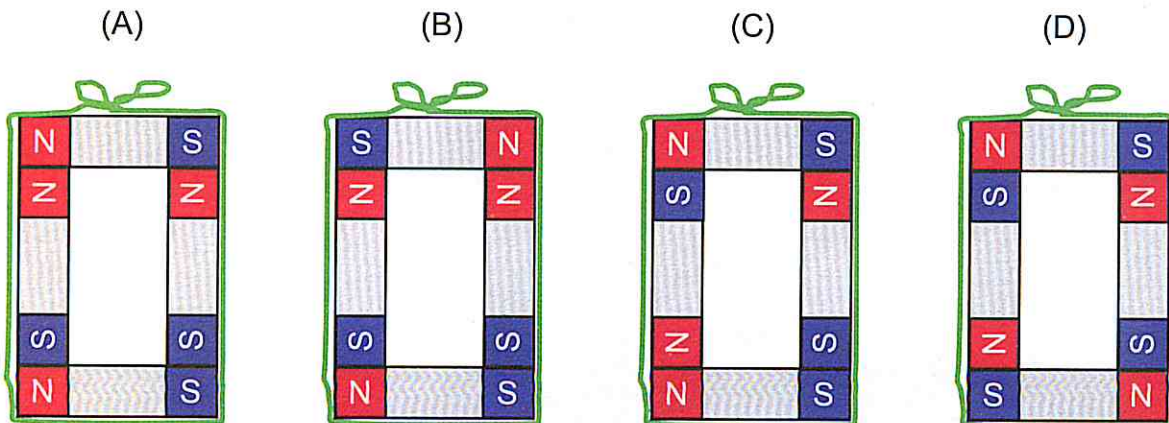
- (A) fibula and talus (B) tibia and fibula
(C) fibula and heel bone (D) tibia and heel bone

11. Magnets have a north end (N) and a south end (S). When two magnets are brought together end to end, they push or pull each other depending on which ends are together. The diagram shows what happens.

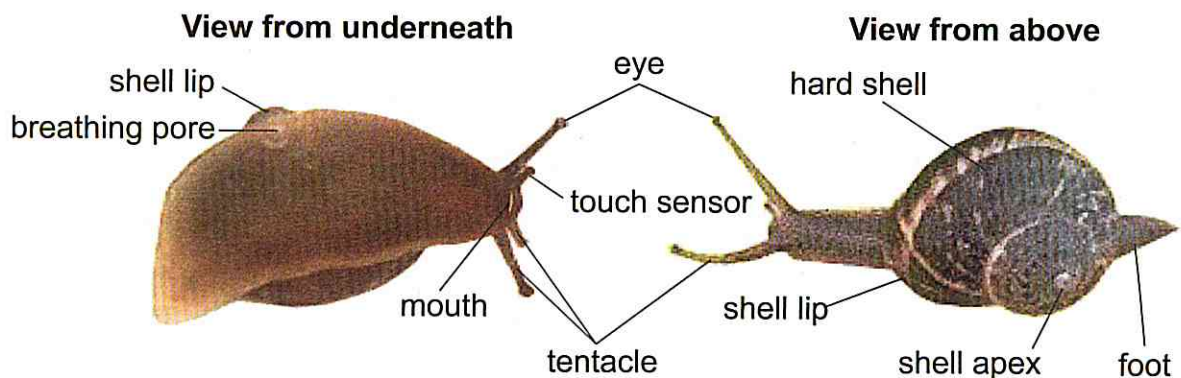


Here are some groups of magnets tied together by string.

Which group will stay together when the string is taken away?



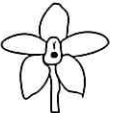
12. The photographs show two views of a garden snail.



From the photographs, which statement is most likely to be true about snails?

- (A) Snails breathe through their mouths.
- (B) Snails use eyes on their two long tentacles to see.
- (C) Snails eat only when they are inside their shells.
- (D) Snails have a shell that spirals from the foot to the shell apex.

13. Orchids are flowering plants. Kim drew this chart to show when five different types of orchids flowered in the bushland near her house.

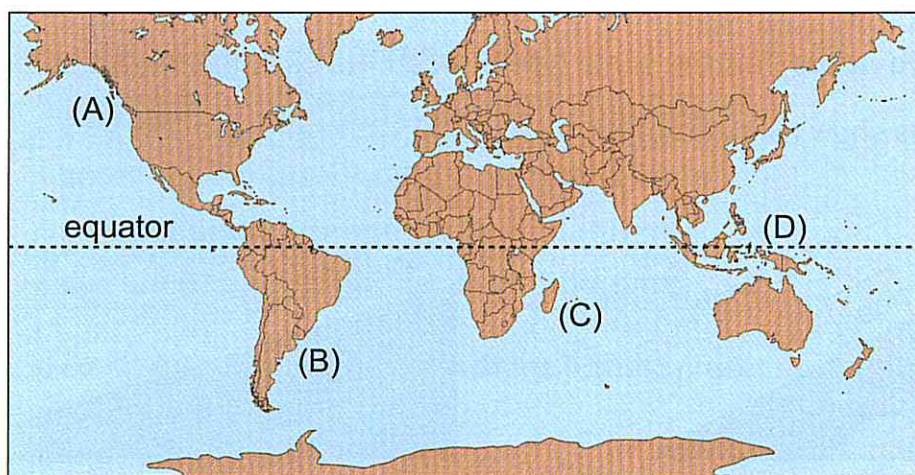
Name	Flower	September	October	November
Greenhood				
Waxlip				
Donkey				
Spider				
Bird				

Which three types of orchids flowered together for the longest time?

- (A) Waxlip, Donkey and Spider
 (B) Waxlip, Donkey and Bird
 (C) Greenhood, Donkey and Waxlip
 (D) Greenhood, Donkey and Bird

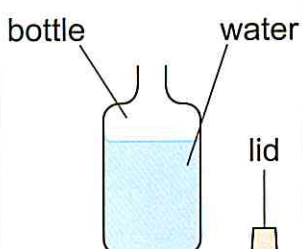
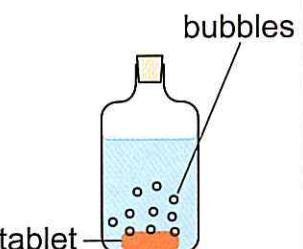
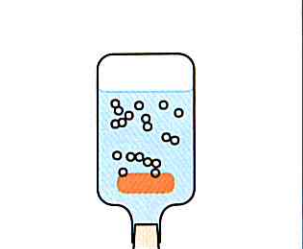
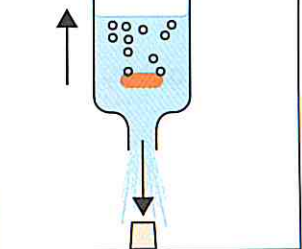
14. Tropical cyclones form only in regions where the ocean water is warm and the air is warm.

Which point on this map shows where tropical cyclones are most likely to form?



For questions 15 and 16 use the information below.

The diagrams show how Annie made a rocket out of a clear plastic bottle.

Some water was poured into the clear plastic bottle.	A tablet was added to the water. Bubbles started to form. The lid was placed on the bottle.	The bottle was turned upside down. More bubbles of gas formed.	The water was pushed out. The bottle moved up.
			

When a chemical reaction occurs, at least one new substance is formed.

15. Which of the following shows that a chemical reaction occurred?

- | | |
|----------------------|---------------------|
| (A) a gas formed | (B) a solid formed |
| (C) the bottle broke | (D) the water froze |

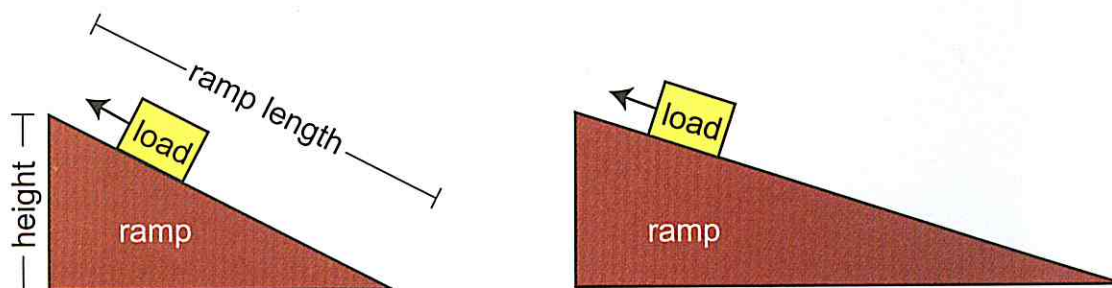
16. The clear plastic bottle that Annie used to make her rocket holds 32 mL of water when completely filled.

Approximately how much water did Annie pour into the plastic bottle?

- | | | | |
|----------|-----------|-----------|-----------|
| (A) 8 mL | (B) 16 mL | (C) 24 mL | (D) 30 mL |
|----------|-----------|-----------|-----------|

17. Tim made two ramps from the same type of wood. He used them to pull a load up a slope.

The diagrams show his ramps.

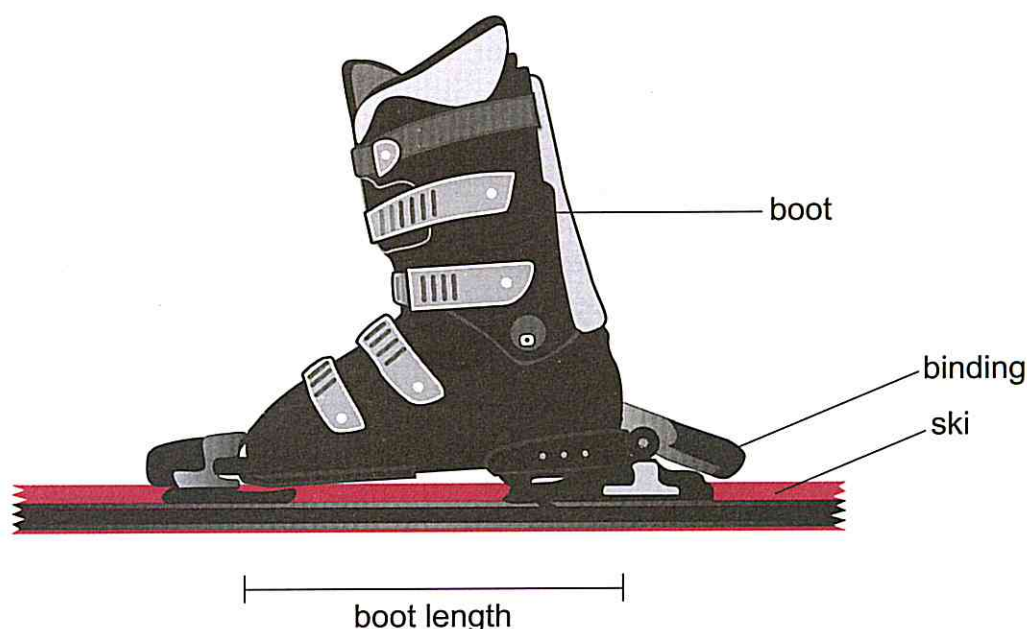


What did Tim change in his investigation?

- | |
|--|
| (A) the smoothness of the ramp surface |
| (B) the height the load was raised |
| (C) the steepness of the ramp |
| (D) the size of the load |

For questions 18 and 19 use the information below.

The picture shows a ski boot connected to a ski at one sixth of its actual size.



18. What is the actual length of the ski boot in centimetres?

- (A) 5 (B) 11 (C) 24 (D) 30

19. The binding connects the boot to the ski. The binding automatically releases the boot if the skier falls.

The boot is released when the pressure on the binding reaches a preset value. The preset release value depends on the skier's mass and skill level. The higher the preset release value, the harder it is to release the boot from the binding.

The table below describes the skill levels of skiers.

Skill level	Skill description
beginner	can ski at low speeds only; turns with difficulty
average	can ski at moderate speeds; turns easily
expert	can ski at high speeds; can make tight turns

Which skier is most likely to use bindings with the highest preset release value?

Skier	Skill level	Mass (kg)
(A)	beginner	45
(B)	expert	65
(C)	expert	85
(D)	average	70

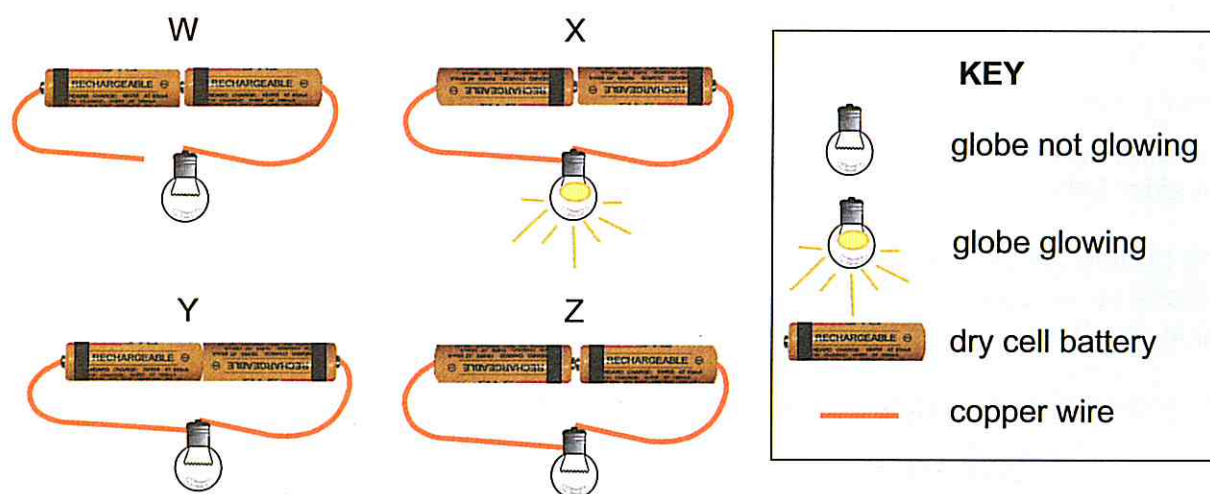
20. Conrad followed these steps to learn about sugar dissolving in water. He used two glass jars of the same size in his experiment.

1. Pour a cup of cold water into one jar.
2. Pour a cup of hot water into the other jar.
3. Place six sugar cubes in each jar.
4. Put the lid on each jar.
5. Leave the jars on the bench.
6. Check each jar every minute for a total of 5 minutes.

The idea Conrad was testing in his experiment was that the time sugar takes to dissolve depends on the

- (A) number of sugar cubes used.
- (B) amount of water used.
- (C) water temperature.
- (D) size of the jar.

21. Vanessa was investigating how she could make an electrical circuit using two batteries, some wire and a globe. She used four different arrangements, W, X, Y and Z.



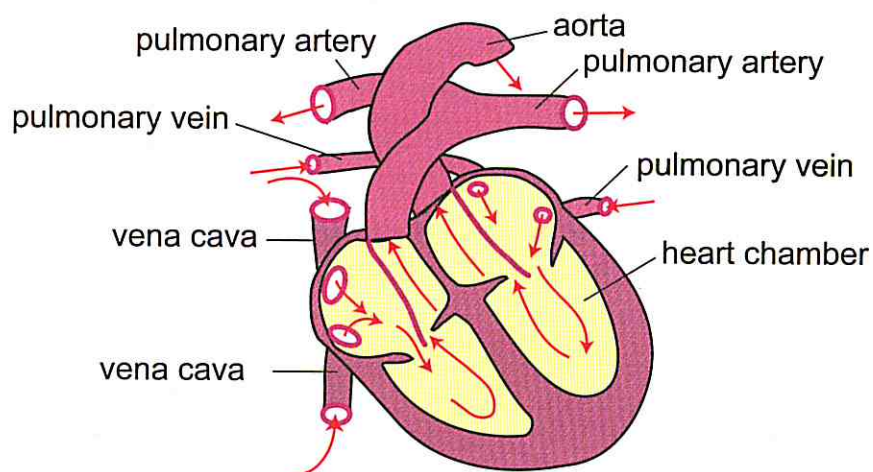
Only arrangement X made the globe light up.

She concluded that the globe will glow only when the batteries are facing in

- (A) the same direction in the circuit and the circuit is open.
- (B) the same direction in the circuit and the circuit is closed.
- (C) opposite directions in the circuit and the circuit is open.
- (D) opposite directions in the circuit and the circuit is closed.

22. Blood vessels carry blood around the body. Arteries carry blood away from the heart. Veins carry blood back to the heart.

The diagram shows parts of the heart and some attached blood vessels. The arrows show the direction of the blood flow.

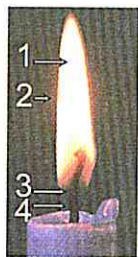


What type of blood vessel is the aorta, and what type is the vena cava?

	Aorta	Vena cava
(A)	vein	vein
(B)	vein	artery
(C)	artery	artery
(D)	artery	vein

23. The photograph shows the zones that occur in the flame of a burning candle.

The table gives information about these zones.



Zone	Colour of light produced	Light is produced by
1	yellow	glowing particles of soot
2	red	fuel burning with oxygen
3	-	heat only, no oxygen present
4	blue	gassy fuel from candle

When fuel from a candle combines with oxygen in the air, it forms carbon dioxide and water.

What is the colour of the light produced by this process?

- (A) red (B) blue (C) colourless (D) yellow

24. Soil contains a number of materials including sand, silt and clay.

The table gives some information about these three materials.

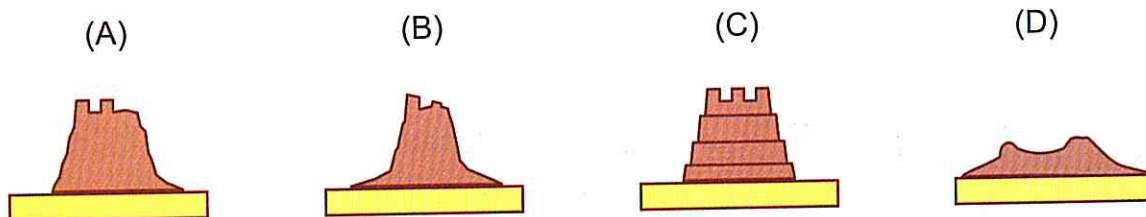
Material	Material holds water	Material sticks together
sand	poorly	weakly
silt	fairly well	well
clay	well	strongly

Jan added different amounts of these materials to some water to make four different mixtures. The table shows how each mixture is made up.

Mixture	Part of mixture		
	Sand	Silt	Clay
1	0	$\frac{1}{4}$	$\frac{3}{4}$
2	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
3	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
4	$\frac{3}{4}$	$\frac{1}{4}$	0

Jan compared the mixtures by shaping each mixture into a castle. Each castle was made from the same amount of mixture. She used the same bucket to shape each castle.

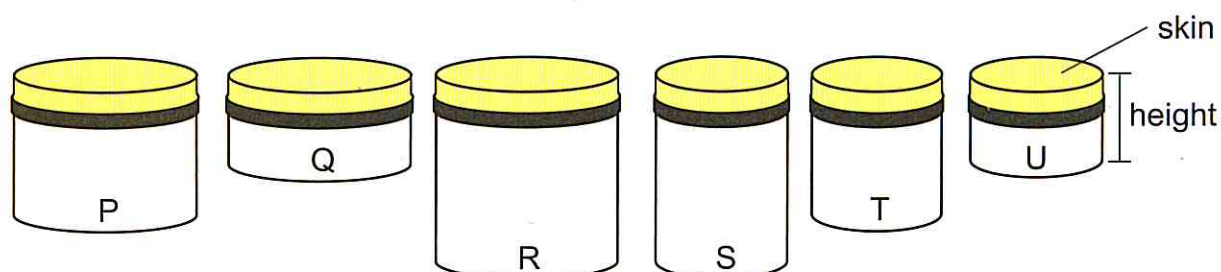
Which castle did Jan most likely make using mixture 2?



25. The pitch of a sound is how high or how low it is.

Julie wants to investigate what happens to the pitch of a drum when she changes the height of the drum, and when she changes the area of the skin at the top of the drum.

The diagram shows some drums.



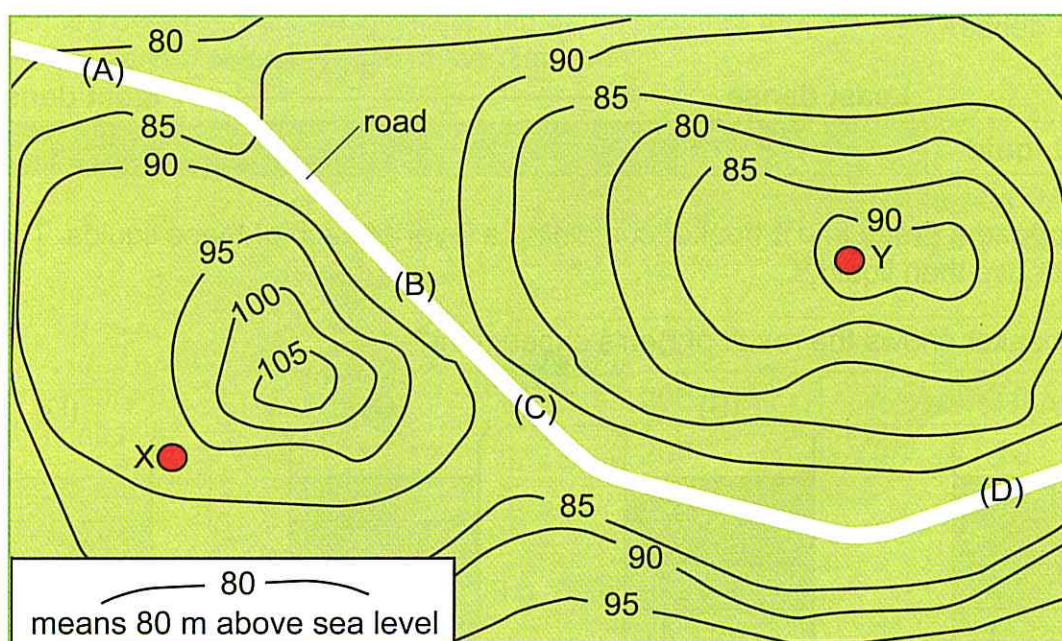
Which drums should Julie use in her investigation?

	Changing the height	Changing the area of the skin
(A)	P and Q	Q and T
(B)	Q and R	P and U
(C)	R and S	P and T
(D)	T and U	Q and U

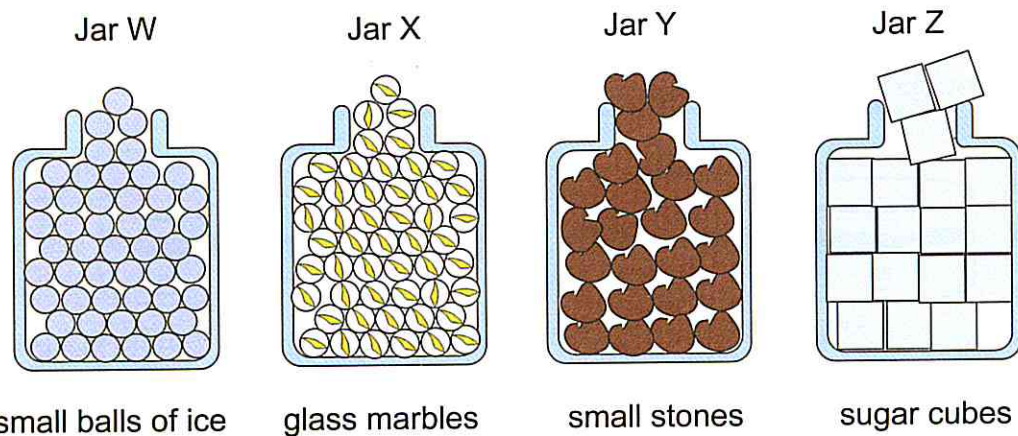
26. Contour lines on a map join places that are the same height above sea level.

The map shows the contour lines for an area with a road passing through it. There are two towers, each ten metres high. One of the towers is located at X and the other at Y.

From which place along the road is only one of the towers visible?



27. Peter filled four glass jars of the same size with different objects. He noticed that although the glass jars were full he could still observe some space inside the jars as shown.



The **next** day Peter could add more objects to only one of the jars without the added objects spilling over.

Which jar was that most likely to be?

- (A) Jar W
- (B) Jar X
- (C) Jar Y
- (D) Jar Z

28. Different substances have different densities.

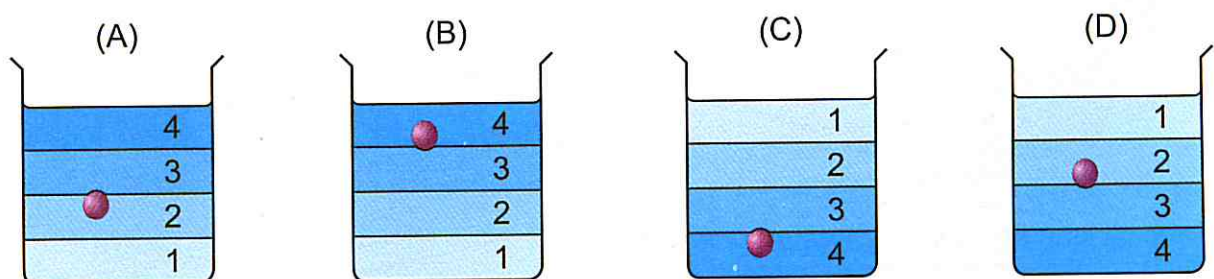
A less dense liquid will float on a liquid that is more dense. A solid will sink in a liquid that is less dense than the solid, and float on a liquid that is more dense.

Ben had four liquids with different densities. The table shows the liquids from least dense to most dense.

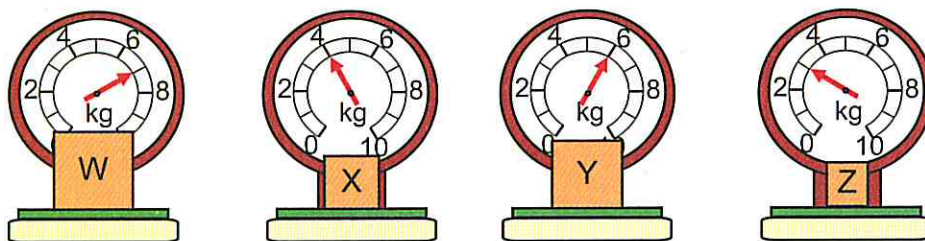
	Least dense	←————→			Most dense
Liquid	1	2	3	4	

Ben dropped a grape into a beaker containing a layer of each of these liquids. The grape is less dense than liquid 3.

Which beaker shows the result of Ben's experiment?

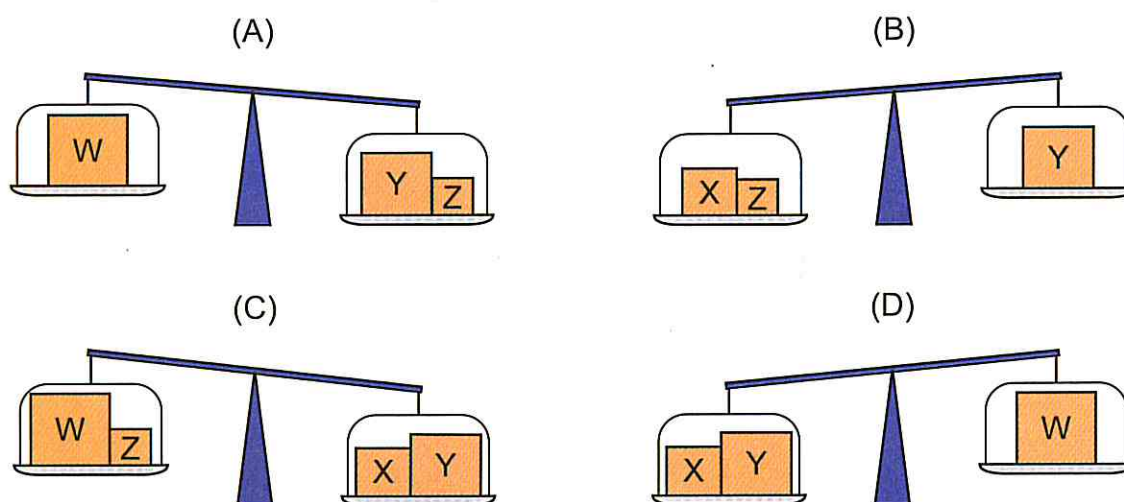


29. Frans used scales to find the mass of each of four different objects.



He then compared the masses of the objects using a balance.

Which diagram shows the balance comparison that is **NOT** correct?

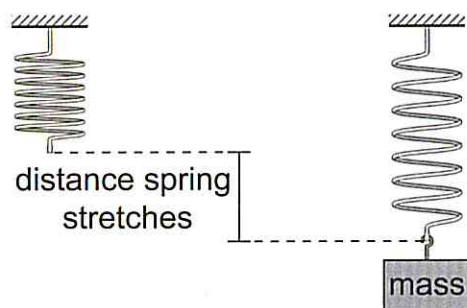


30. When a mass is attached to a spring, the spring stretches.

Michelle has a spring that is 200 mm long with no mass attached.

She attaches different masses to this spring and measures the distance that the spring stretches for each mass as shown in the diagram.

The table shows her results.



Attached mass (g)	Distance that the spring stretches (mm)
0	0
100	4
200	8
300	12
400	16

How long is the spring with a mass of 600 g attached?

- (A) 20 mm (B) 24 mm (C) 220 mm (D) 224 mm

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The following year levels should sit **THIS Paper**:

Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific	Year 4
Singapore	Primary 3
South Africa	Grade 4



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